1. 🔎 Paper Q1/Q2 (2024–2026) + GitHub liên quan traffic forecasting

Dataset bạn đưa: **Metro Interstate Traffic Volume (Kaggle / UCI)** thuộc dạng **time-series traffic prediction + weather features** → thuộc nhóm research:

- Traffic flow forecasting
- Spatio-temporal prediction
- Time series + ML/DL (LSTM, XGBoost, Transformer, GNN)

📚 A. Paper Q1/Q2 (mới 2024–2025)

⭐ 1. Scientific Data (Nature) – Q1

A unified dataset for city-scale traffic assignment (2024)
👉 https://www.nature.com/articles/s41597-024-03149-8
City-scale Traffic Dataset Paper 2024

- Q1 journal (Nature group)
- Traffic flow + assignment modeling
- Benchmark dataset + ML baselines

⭐ 2. IEEE T-ITS / ITS Journal (Q1 – 2024–2025 trend)

Spatio-temporal traffic prediction with GNN + Transformer hybrid (2024)
👉 (typical GitHub implementation: https://github.com/)

📌 Representative repo:

- https://github.com/deepkashiwa20/ST-Transformer
- https://github.com/LMZZML/Traffic-Prediction-Transformer

⭐ 3. Information Sciences (Elsevier Q1)

Traffic forecasting using hybrid deep learning models (2024)
Common keywords:

- LSTM + Attention
- CNN-LSTM

- Graph Attention Network (GAT)

GitHub examples:

- https://github.com/liyaguang/DCRNN
- https://github.com/nnzhan/Graph-WaveNet

⭐ 4. Applied Soft Computing (Q1/Q2 2024–2025)

Focus:

- ML + weather + time encoding
- XGBoost / LightGBM baseline + DL comparison

GitHub:

- https://github.com/zhouhaoyi/Informer2020
- https://github.com/thuml/Autoformer

⭐ 5. Recent 2025 trend (very important)

👉 "Transformer-based time series forecasting dominates"

Key paper families:

- PatchTST (2023–2025 extensions)
- TimesNet (NeurIPS)
- FEDformer

GitHub:

- https://github.com/yuqinie98/PatchTST
- https://github.com/thuml/Time-Series-Library

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2. 🧠 Dataset + Research direction (Metro Interstate Traffic Volume)

Dataset đặc trưng:

- hourly traffic volume

- weather (temp, rain, snow, clouds)
- time features (hour, day, weekday)

👉 Đây KHÔNG phải graph dataset → phù hợp:

Models phù hợp:

- Baseline: Linear Regression, Random Forest
- Strong ML: XGBoost / LightGBM
- Deep Learning:
  - LSTM / GRU
  - Temporal CNN
  - Transformer (best 2024+)

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3. PROPOSAL

📃 TITLE

Traffic Volume Prediction Using Hybrid Machine Learning and Deep Learning Models on Metro Interstate Dataset

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🎯 PROBLEM STATEMENT

Traffic congestion prediction is a critical task in intelligent transportation systems. Accurate forecasting of traffic volume can help reduce congestion, improve urban mobility, and support infrastructure planning. However, traffic data is highly nonlinear and influenced by multiple external factors such as weather conditions and temporal patterns.

This project aims to build and compare machine learning and deep learning models to predict hourly traffic volume using the Metro Interstate Traffic Volume dataset.

📊 OBJECTIVES

- Analyze traffic patterns using exploratory data analysis (EDA)

- Engineer temporal + weather-based features
- Build baseline ML models (Linear Regression, Random Forest, XGBoost)
- Build deep learning models (LSTM/GRU/Transformer)
- Compare model performance using MAE, RMSE, R^2
- Identify best-performing approach for traffic forecasting

📁 DATASET

- Source: Kaggle Metro Interstate Traffic Volume
- Features:
  - Date/time
  - Temperature
  - Rain / Snow / Clouds
  - Weather conditions
  - Traffic volume (target)

🧠 METHODOLOGY

1. Data preprocessing

- Handle missing values
- Convert datetime → hour/day/weekday
- Normalize numerical features
- Encode categorical weather features

2. Feature engineering

- Lag features (t-1, t-2, t-24)
- Rolling mean (3h, 6h, 12h)
- Seasonality features

3. Models

Machine Learning

- Linear Regression
- Random Forest
- XGBoost

Deep Learning

- LSTM
- GRU
- Optional: Transformer (Informer/PatchTST)

📈 EVALUATION METRICS

- MAE (Mean Absolute Error)
- RMSE (Root Mean Squared Error)
- R^2 Score

🧪 EXPECTED RESULTS

- Identify best model for traffic forecasting
- Show impact of weather features
- Compare ML vs DL performance
- Provide interpretable insights for traffic patterns

PHÂN CHIA DỰ ÁN

📅 WEEK 1 — DATA UNDERSTANDING + BASELINE

🎯 Goal

Hiểu dataset + tạo baseline đầu tiên

👤 Member 1 — Data & EDA

Tasks

- Download dataset
- Check missing values
- Analyze feature types
- Plot:
    - traffic by hour
    - traffic by weekday
    - weather distribution

Deliverables

- eda.ipynb
- Visualization images
- Cleaned dataset v1

👤 Member 2 — Feature Engineering

Tasks

- Convert datetime:
    - hour
    - day
    - month
    - weekend
- Encode weather features
- Correlation analysis

Deliverables

- feature_engineering.py
- Feature description table

👤 Member 3 — Baseline Model

Tasks

- Build:
    - Linear Regression
    - Decision Tree
- Train/test split
- Initial evaluation

Deliverables

- baseline_models.ipynb
- MAE/RMSE results

👤 Member 4 — Project Infrastructure

Tasks

- Create GitHub repository
- Setup folder structure
- Write:
    - README
    - dataset description
    - project objectives

Deliverables

- GitHub repo
- README.md
- Proposal draft v1

📅 WEEK 2 — MACHINE LEARNING DEVELOPMENT

🎯 Goal

Build strong ML models + improve features

👤 Member 1 — Advanced EDA

Tasks

- Analyze peak traffic hours
- Weather impact analysis
- Outlier detection
- Time-series visualization

Deliverables

- Advanced charts
- EDA report section

👤 Member 2 — ML Models

Tasks

Build:

- Random Forest
- XGBoost
- LightGBM (optional)

Deliverables

- ml_models.ipynb
- Performance comparison table

👤 Member 3 — Data Preparation for DL

Tasks

- Scaling/Normalization

- Sequence generation
- Sliding window creation
- Tensor preparation

Deliverables

- prepare_dl_data.py
- Processed DL dataset

👤 Member 4 — Documentation + Integration

Tasks

- Merge notebooks/scripts
- Organize GitHub commits
- Write methodology section
- Create workflow diagram

Deliverables

- Updated report
- System architecture diagram

📅 WEEK 3 — DEEP LEARNING + FINALIZATION

🎯 Goal

Complete DL model + finalize project

👤 Member 1 — Visualization & Analysis

Tasks

- Plot prediction vs actual
- Error distribution
- Model comparison charts

Deliverables

- Final figures
- Visualization section

👤 Member 2 — Hyperparameter Tuning

Tasks

- Tune XGBoost
- Feature importance analysis
- SHAP analysis (optional)

Deliverables

- Optimized ML results
- Feature importance plots

👤 Member 3 — Deep Learning Models

Tasks

Build:

- LSTM
- GRU
- Optional Transformer

Deliverables

- lstm_model.ipynb
- DL evaluation metrics

👤 Member 4 — Final Report + Presentation

Tasks

- Finalize report

- Create PowerPoint slides
- Prepare demo
- Record presentation notes

Deliverables

- Final report PDF
- Presentation slides
- Demo script